4-5 Probability and Counting Rules

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This lesson is where Chapter 4 comes together: the counting rules of §4-4 produce `n(E)` and `n(S)`, and the classical probability formula of §4-1 divides them. Course-outline criterion CLO4-A — *calculate probabilities using counting rules and basic probability principles* — describes exactly this kind of question.

TOOLS CARRIED OVER FROM §4-1 AND §4-4

Classical probability: $P(E) = n(E) / n(S)$

Combination: ${}_nC_r = n! / [(n - r)! \cdot r!]$

Fundamental counting rule: $k_1 \times k_2 \times \cdots \times k_n$

At least one: $P(\text{at least one}) = 1 - P(\text{none})$

Choosing from several categories ("2 defective **and** 2 good") → one combination per category, multiplied together.

1 · The three-step method

This is the destination. The counting rules supply `n(E)` and `n(S)`; classical probability divides them. The structure of every §4-5 question is identical:

STEP 1 **Count the sample space.**
How many total ways can the selection be made? Usually a single ${}_nC_r$ or a product.

STEP 2 **Count the event.**
How many of those ways satisfy the condition? Usually a product of smaller `C`'s — one per category.

STEP 3 **Divide.**

`P(E) = n(E) / n(S)`, then round sensibly and interpret in words.

الشرح بالعربي — الاحتمال باستخدام قواعد العدّ (أهم نوع أسئلة)

كل سؤال من هذا النوع له نفس الخطوات الثلاث:

1. **احسب `n(S)`:** عدد كل طرق الاختيار الممكنة — غالبًا `nCr` واحدة.
2. **احسب `n(E)`:** عدد الطرق التي تحقق الشرط — غالبًا حاصل ضرب توافيق، واحدة لكل فئة.
3. **اقسم:** `P(E) = n(E) ÷ n(S)`، ثم اكتب جملة تفسيرية.

مثال: 24 قطعة منها 4 معيبة، نختار 4. احتمال قطعتين معيبتين بالضبط = `(4C2 × 20C2) ÷ 24C4 = 1140 ÷ 10626 ≈ 0.1073`. وإذا طُلب «واحدة معيبة على الأقل» ← `1 − P(none) = 1 − 0.4560 = 0.5440`.

قاعدة الاتساق: استخدم نفس نوع العدّ في البسط والمقام — لا تضع تباديل فوق توافيق.

EXAMPLE 4-51 · DEFECTIVE INTEGRATED CIRCUITS

A box contains 24 integrated circuits, 4 of which are defective. If 4 are sold at random, find: (a) exactly 2 are defective; (b) none is defective; (c) all are defective; (d) at least 1 is defective.

Set-up. 4 defective, 20 good, 24 total. Choosing 4 with no order → combinations.

$$n(S) = {}_{24}C_4 = \frac{24 \cdot 23 \cdot 22 \cdot 21}{4 \cdot 3 \cdot 2 \cdot 1} = \frac{255{,}024}{24} = \mathbf{10{,}626}$$

1. **(a) Exactly 2 defective** — so 2 from the 4 defective *and* 2 from the 20 good:

   $$n(E) = {}_4C_2 \times {}_{20}C_2 = 6 \times 190 = 1{,}140$$

   $$P = \frac{1{,}140}{10{,}626} \approx \boxed{0.1073}$$

2. **(b) None defective** — all 4 from the 20 good:

   $$n(E) = {}_{20}C_4 = \frac{20 \cdot 19 \cdot 18 \cdot 17}{24} = 4{,}845$$

   $$P = \frac{4{,}845}{10{,}626} \approx \boxed{0.4560}$$

3. **(c) All defective** — all 4 from the 4 defective:

   $$n(E) = {}_4C_4 = 1$$

   $$P = \frac{1}{10{,}626} \approx \boxed{0.000094}$$

4. **(d) At least 1 defective** — the complement of "none defective", which is part (b):

   $$P = 1 - 0.4560 = \boxed{0.5440}$$

PART (D) IS THE EXAM'S REWARD FOR RECOGNISING THE COMPLEMENT

Done the long way, (d) = P(1 defective) + P(2) + P(3) + P(4) — four calculations. Done as `1 − P(none)`, it is one subtraction you have already computed. This is precisely why parts (b) and (d) appear in the same question: the examiner is testing whether you noticed.

EXAMPLE 4-52 · TERM PAPER SELECTION

A student needs to select two topics to write two term papers for a course. There are 8 topics in economics and 11 topics in science. Find the probability that she selects one topic in economics and one topic in science.

1. **n(S):** any 2 of the 19 available topics → $_{19}C_2 = \frac{19 \cdot 18}{2} = 171$
2. **n(E):** one economics *and* one science → $_8C_1 \times {}_{11}C_1 = 8 \times 11 = 88$
3. $P = \frac{88}{171} \approx$ **0.5146**

EXAMPLE 4-53 · STATE LOTTERY NUMBER

In the Pennsylvania State Lottery, a person selects a three-digit number and repetitions are permitted. If a winning number is selected, find the probability that it will have all three digits the same.

1. **n(S):** three digits, repetition permitted → $10 \times 10 \times 10 = 1{,}000$
2. **n(E):** all three digits the same → 000, 111, 222, ..., 999 → 10
3. $P = \frac{10}{1{,}000} =$ **0.01** (1%)

EXAMPLE 4-54 · TENNIS TOURNAMENT

There are 8 married couples in a tennis club. If 1 man and 1 woman are selected at random to plan the summer tournament, find the probability that they are married to each other.

1. **n(S):** 8 men to choose from *and* 8 women → $8 \times 8 = 64$ pairs
2. **n(E):** only 8 of those pairs are actually married to each other (one per couple) → 8
3. $P = \frac{8}{64} = \frac{1}{8} =$ **0.125**

Alternative reasoning that proves the answer: pick any man first; his wife is 1 of the 8 women, so P = 1/8 directly. ✔

HW 11 · SOCKS IN A DRAWER

A drawer contains 11 identical red socks and 8 identical black socks. Suppose you choose 2 socks at random in the dark. (a) What is the probability that you get a pair of red socks? (b) A pair of black socks? (c) 2 unmatched socks?

n(S): any 2 of 19 socks → $_{19}C_2 = \frac{19 \cdot 18}{2} = 171$

1. **(a) Pair of red:** $_{11}C_2 = \frac{11 \cdot 10}{2} = 55 \rightarrow P = \frac{55}{171} \approx$ **0.3216**
2. **(b) Pair of black:** $_{8}C_2 = \frac{8 \cdot 7}{2} = 28 \rightarrow P = \frac{28}{171} \approx$ **0.1637**
3. **(c) 2 unmatched:** one red *and* one black: $_{11}C_1 \times {}_{8}C_1 = 88 \rightarrow P = \frac{88}{171} \approx$ **0.5146**

   Or by complement: 1 − 0.3216 − 0.1637 = 0.5146 ✔

Check: 55 + 28 + 88 = 171 = n(S). The three cases are exhaustive and mutually exclusive, so their counts must total the sample space. ✔

Chapter 4 study series — 11 files. Read the lessons in order; the revision files at the end draw on all of them.

00 · Exam brief — what the outline tests

01 · Lesson 4-1 — Sample Spaces and Probability

02 · Lesson 4-2 — The Addition Rules

03 · Lesson 4-3 — Multiplication Rules and Conditional Probability

04 · Lesson 4-4 — Counting Rules

05 · Lesson 4-5 — Probability and Counting Rules ← this file

06 · Master formula sheet

07 · Which rule do I use?

08 · The 14 traps that cost marks

09 · Review exercises, solved

10 · The A+ study protocol

Rebuilt from Dr. Ahmed Al-Hadhrami's lecture slides (Spring 24) and aligned to the SQU College of Science STAT 2101 course outline, Fall 2026. Textbook: Allan G. Bluman, *Elementary Statistics: A Step by Step Approach*, brief version, 10th edition, McGraw Hill (2017). A companion 100-question bank with full solutions accompanies this series.